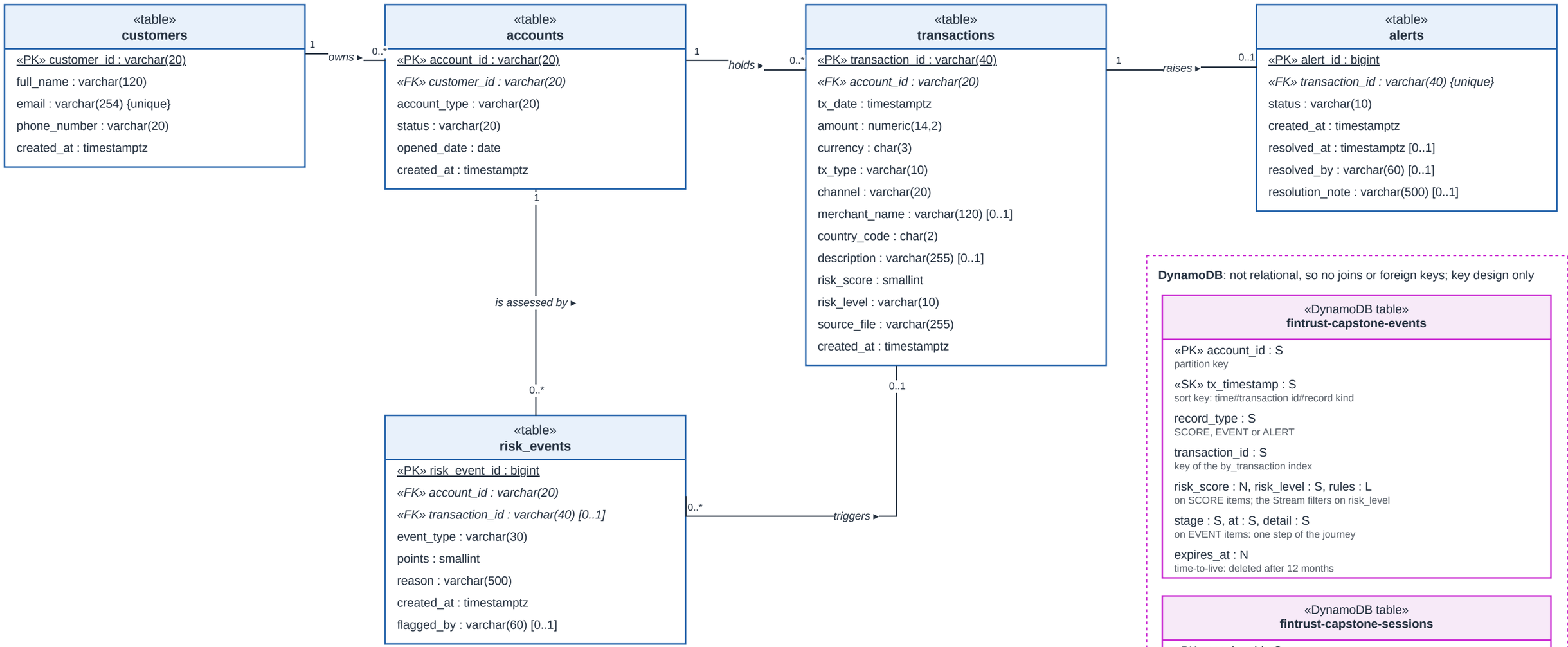
FinTrust ledger: entity relationship diagram (UML class notation)

PostgreSQL 16 on Amazon RDS · third normal form · generated from docs/erd.md, which a test compares with the deployed schema



DynamoDB: not relational, so no joins or foreign keys; key design only

«DynamoDB table»
fintrust-capstone-events

- «PK» account_id : S
  partition key
- «SK» tx_timestamp : S
  sort key: time#transaction id#record kind
- record_type : S
  SCORE, EVENT or ALERT
- transaction_id : S
  key of the by_transaction index
- risk_score : N, risk_level : S, rules : L
  on SCORE items; the Stream filters on risk_level
- stage : S, at : S, detail : S
  on EVENT items: one step of the journey
- expires_at : N
  time-to-live: deleted after 12 months

«DynamoDB table»
fintrust-capstone-sessions

- «PK» session_id : S
  the sign-in's origin_jti from Cognito
- username : S, display_name : S, roles : L
  who, and which role
- status : S, end_reason : S
  ACTIVE or ENDED, and why it ended
- started_epoch, last_seen_epoch, ends_epoch : N
  the 15-minute idle and 8-hour limits
- user_agent : S
  which browser signed in
- expires_at : N
  time-to-live: deleted after 90 days

How to read it

«table»	one PostgreSQL table; each line is a column, as name : type
«PK» underlined	primary key: identifies each row
«FK» italic	foreign key: refers to the primary key of the table it is linked to
[0..1]	the column may be empty (NULL); every other column is NOT NULL
{unique}	no two rows may hold the same value
1 · 0..1 · 0..*	multiplicity at each end: exactly one, zero or one, zero or more
owns ▸	the relationship's name, read in the direction of the arrowhead

Rules the diagram cannot show

CHECK constraints on every type, channel, level and status · amount > 0 · one alert per transaction · a resolved alert records who and when · only a manual flag names a person · ON DELETE RESTRICT on every foreign key.

Indexes

transactions (account_id), (tx_date), (account_id, tx_date), (risk_level) · accounts (customer_id) · risk_events (account_id), (transaction_id) · alerts (status)